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Simulation of oxytactic microbes in hybrid nanofluid with activation energy and LTNE effects using Bayesian regularization neural network technique

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Abstract

This model has significant applications in advanced thermal and bioengineering systems where coupled fluid flow, heat transport, and microbial dynamics are vital. It is applicable to the design of bioreactors and microbial fuel cells, where Oxytactic bacteria improve mixing and mass transfer efficiency. The model optimizes heat transport in nanofluid-based cooling systems, particularly under magnetic fields (MHD), by employing the Xue thermal conductivity formulation. The inclusion of LTNE (local thermal non-equilibrium) effects and activation energy makes it applicable to chemical reactors, catalytic processes, and biomedical drug delivery, all of which rely on reaction rates and temperature gradients. This study used a neural network trained using the Bayesian-Regularized approach to inspect the impact of LTNE on oxytactic microorganisms in bioconvection flow of a HNF (hybrid nanofluid) with activation energy. There are several uses for Oxytactic microbes, especially in environments and industries. In wastewater treatment, oxytactic bacteria are frequently utilized to increase oxygen distribution in bioreactors and accelerators for the breakdown of organic contaminants. These bacteria move quickly in response to gradients in oxygen concentration. In many biotechnological applications, Oxytactic microbes must be able to situate themselves within fluid flows in order to efficiently combine and transfer nutrients. The concentration profile increases as the activation energy parameter's values rise.

Keywords Oxytactic microorganism, LTNE effects, Artificial neural network, Hybrid nanofluid, Activation energy

1 Introduction

A machine learning method called an artificial neural network (ANN) is intended to replicate the composition and operations of the human brain. Layers of neurons, which are interconnected processing units, make up the system. External data is collected by the input layer and sent to the buried levels. The output layer receives the outcomes of the complex computations performed on the input data by the hidden layers. The output



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layer generates the network's ultimate output. By using fast, data-driven models rather than expensive numerical simulations, artificial neural networks can be utilized to derive parameters without solving all of the governing equations, improve heat transfer efficiency, and lower computational costs. The BRT (Bayesian regularization technique) was used in a machine learning research by Shaaban et al. [1] to examine the bioconvection flow of Boger NF (nanofluid) in the presence of oxytactic bacteria. Quresh [2] models the properties of heat transmission on an MHD Casson fluid film flow across a surface using an ANN simulation. The convective flow of THNF (ternary hybrid nanofluid) with and LTNE using an ANN improved using the BR (Bayesian-regularization) approach were inspected by Abbas et al. [3]. Alzahrani et al. [4] inspect the prediction of MHD natural convection in THNF using neural networks. Hussein et al. [5] used Cattaneo-Christov theory to inspect the ANN design for DFF (Darcy-Forchheimer flow) of a Boger HNF (hybrid nanofluids). Using a linear regression approach, Abdal et al. [6] investigate a ML (machine learning) research for MHD Casson fluid flow with CCHF. The machine learning simulation of unstable convection in micropolar fluids with Lorentz properties were scrutinized by Mahboobtosi et al. [7]. Rout and Nandkeolyar [8] investigate the Hall current and electric field impacts on magnetohydrodynamic THNF flow across a sheet using numerical and neural network analysis.

Hybrid nanofluids, an improved type of thermal fluid, are created when two or more dissimilar NPs (nanoparticles) float in a basic fluid. This unique strategy takes benefit of the particular idiosyncrasies of distinct NPs kinds to increase the temperature properties and constancy of the liquid. Thermal convergence, specific heat, viscosity, and convective heat transfer coefficients and speeds may all be improved by the interactions among the particles as compared with simple nanofluids made up of a single nanoparticle. Its usage in cooling circuits for electric and electronics components is one of its best uses since temperature management is crucial for both operation and longevity. The temperature conductivity of cooling systems can be significantly increased by new hybrid nanofluids, enabling the development of smaller but more potent electronics. Hafeez et al. [9] examined how heat transmission affects the MHD flow of a radiative HNF with a wedge. Rehman et al. [10] examined the MHD movement of blood-based HNF counting CNTs under the result of thermal radiation on a permeable sheet. Mumtaz et al. [11] inspected the thermal conductivity of quaternary HNF with radiative and Joule heating properties in MHD flow on a surface. A numerical research of magnetohydrodynamic HNF flow with heat transmission under temperature radiation among two coaxial cylinders is conducted by Bilal et al. [12]. The two-phase modeling of micropolar NF flow on a surface were studied by Nayak et al. [13] using chemical reaction analysis and Darcy–Forchheimer inertia. Farooq et al. [14] inspected MHD natural convection of HNF in a porous wavy trapezoidal cavity using finite element and NNT. Using a hybrid approach, Nayak et al. [15] inspected the behavior of viscoelastic fluid flow across a permeable media with a heat generation. The natural convection flow of HNF in a triangular inclusion with a heated wavy wall and a cold impediment was inspected by Turabi et al. [16]. The properties of heat dissipation and Darcy-Forchheimer inertial drag on micropolar HNF flow via mounting surface were scrutinized by Nayak et al. [17]. Using the finite element method, Shafique et al. [18] scrutinized the numerical simulation and mathematical modelling of blood flow in stenotic arteries. The computational investigation of heat transmission for HNF flow in a wavy lid-driven cavity with heating and entropy formation was

scrutinized by Malik et al. [19]. The Cattaneo–Christov mass and heat flux on micro-polar NF with chemical reaction and inertial drag were scrutinized by Nayak et al. [20]. Entropy formation and natural convection in a wavy triangular cavity with Casson fluid under radiation and MHD were considered numerically and by optimization analysis by Turabi et al. [21].

Thermal transport processes in permeable medium have been widely studied using LTNE and LTE methods. These two methods are used to inspect the temperature at the fluid-porous zone interaction. The LTNE approach uses two different energy equations to link the porous media's fluid and solid phases, but the LTE model neglects the temperature differential between them. Geometric and thermos-hydraulic properties must be taken into deliberation while selecting these approaches, because the LTNE approach requires more dispensation power than the LTE method. The mixed convective nanoliquid flow across a wedge was investigated by Patil and Shankar [22] using an LTNE model. Sarada et al. [23] inspected the properties of MHD on the heat transmission effect of fluid flow over a surface under LTNECs. Mokhtari and Tayebi [24] study second law investigation and magneto-thermosolutal natural convection in a NF-filled enclosure protected by a non-Darcian porous wall. They also examine Joule heating and heat generation properties. The thermosolutal-assisted free convection in a non-Darcy porous annulus filled with NF was inspected by Lahrech et al. [25] using an LTNE method. Siddabasappa et al. [26] scrutinized the LTNE approach to study the MHD flow via a permeable channel and heat transmission of a Newtonian.

The activation energy is the low energy needed to start a certain chemical reaction. The chemical processes and mass transport often interact in complex ways. It may be confirmed by using liquid flow and reactant species digestion and fabrication at different mass transfer speeds. For the first time, Arrhenius proposed the Arrhenius AE (activation energy) for classifying chemical processes according to species. The mobility of nanoparticles in simple carrier fluids is also significantly influenced by the AE value. The effects of AE on the transfer of mass and heat in MHD flow of THNF across permeable discs with different geometries were scrutinized by Wantao et al. [27]. Goud et al. [28] used AE and joule heating features to study the heat transmission features of MHD temperature radiation flowing through a sheet. The AE and chemical reaction effects were among the temperature and MHD properties of pure and HNF was scrutinized by Fatima et al. [29]. Ahmed et al. [30] inspected the numerical modeling of an MHD nonlinear radiative Maxwell nanofluid with AE. The consequence of heat production and molecule AE on the heat transmission of radiative NF flow over a surface was analysed by Kattaswamy et al. [31].

Examining the impact of LTNE effects on Oxytactic microbes in hybrid nanofluid flow with activation energy characteristics is the aim of this research. The goal of the study is:

- To examine how Oxytactic microbes effect in HNF using Xue thermal conductivity model.
- The Bayesian-Regularized network-based computing solver for thorough simulation analyses of the fluidic system is validated by the convergence profiles of the mean squared errors (MSEs), the adaptive behavior of the optimization algorithm's control parameters, the error-based histograms, and the regression statistics.
- To assess how the heat source affects the thermal transportation of the hybrid nanofluid.

- To examine the effects of the activation energy on thermal boundary layer flow of hybrid nanofluid.
- To inspect the influence of MHD on the bioconvection flow.

2 Mathematical formulations

The significance of LTNE and heat generation effects on Oxytactic microbes in two-dimensional, incompressible, electrically conducting HNF (hybrid nanofluid) flow across a sheet with Arrhenius activation energy and thermal radiation using Xue model. Figure 1 illustrates how the flow starts on the plate at $y = 0$. The flow assumptions as shown below.

- The fluid is a hybrid nanofluid made up of *Mgo* and *Ag* in water, which improves thermal conductivity and heat transmission efficiency.
- Lorentz forces are created when a uniform MF (magnetic field) is applied perpendicular to the sheet. These forces resist fluid motion, lowering velocity and influencing momentum transmission.
- The thickness of the TBL (thermal boundary layer) is greatly augmented by thermal radiation, which increases heat transfer.
- The Arrhenius activation energy affects mass transfer processes, species concentration, and the pace of chemical reactions.

The conservation of momentum, concentration, mass, oxytactic microbes, liquid and solid phase thermal heat equations for HNF are define as follows [27, 32–34]:

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 0, \quad (1)$$

$$\rho_{Thnf} \left(Q \frac{\partial P}{\partial y} + P \frac{\partial P}{\partial x} \right) = \mu_{hnf} \frac{\partial^2 P}{\partial y^2} - \frac{\mu_{hnf}}{k} P - \sigma_{hnf} B_0^2 P \quad (2)$$

$$(\rho c_p)_{hnf} \left(P \frac{\partial R}{\partial x} + Q \frac{\partial R}{\partial y} \right) = (k_{hnf} + \frac{16\sigma^* R_\infty^3}{3k^*}) \frac{\partial^2 R}{\partial y^2} + h_{fs} (R_p - R) - A_0 (R - R_\infty), \quad (3)$$

$$0 = \frac{k_p}{(\rho c_p)_p} \left(\frac{\partial^2 R_p}{\partial y^2} \right) + \frac{h_{fs}}{(\rho c_p)_p} (R - R_p), \quad (4)$$

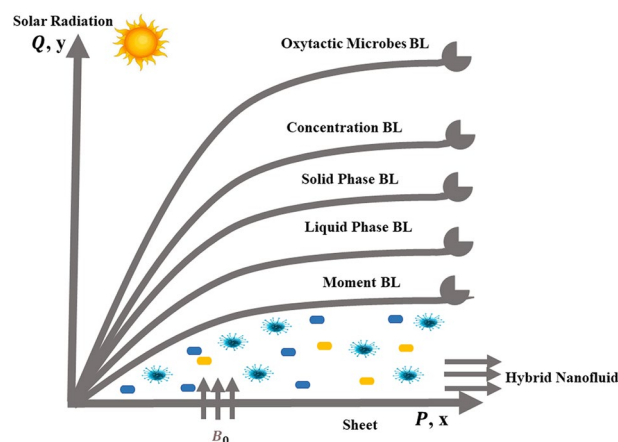


Fig. 1 Flow configuration

$$P \frac{\partial S}{\partial x} + Q \frac{\partial S}{\partial y} = D_B \frac{\partial^2 S}{\partial y^2} - k_r^2 (S - S_\infty) \left(\frac{S}{R_\infty} \right)^m \exp\left(\frac{-Ea}{KR}\right), \quad (5)$$

$$P \frac{\partial G}{\partial x} + Q \frac{\partial G}{\partial y} = D_m \frac{\partial^2 G}{\partial y^2} - \frac{bW_m}{B_w - B_\infty} \left[\frac{\partial}{\partial y} \left(G \frac{\partial B}{\partial y} \right) \right], \quad (6)$$

$$P \frac{\partial B}{\partial x} + Q \frac{\partial B}{\partial y} = D_s \frac{\partial^2 B}{\partial y^2} - \frac{\gamma (G - G_\infty)}{\Delta B}. \quad (7)$$

Boundary conditions [27, 32–34]:

$$P = P_w, R = R_w, B = B_w \text{ at } y = 0, \quad (8)$$

$$S = S_w, Q = 0, R_p = r_w, G = G_w, \text{ at } y = 0, \quad (9)$$

$$P \rightarrow 0, S \rightarrow S_\infty, G \rightarrow G_\infty, \text{ as } y \rightarrow \infty, \quad (10)$$

$$R \rightarrow R_\infty, R_p \rightarrow r_\infty, B \rightarrow B_\infty, \text{ as } y \rightarrow \infty, \quad (11)$$

$$P = \frac{\partial \psi}{\partial y}, \psi(x, y) = \nu_f X f'(\xi), Q = -\frac{\partial \psi}{\partial x}, \quad (12)$$

$$\xi = \frac{y}{L}, X = \frac{x}{L}, \quad (13)$$

$$\Theta(\xi) = \frac{G - G_\infty}{G_w - G_\infty}, \theta_p(\xi) = \frac{R_p - r_\infty}{r_w - r_\infty}, \quad (14)$$

$$\begin{aligned} \theta(\xi) &= \frac{R - R_\infty}{R_w - R_\infty}, \\ \phi(\xi) &= \frac{S - S_\infty}{S_w - S_\infty}, \\ \Psi(\xi) &= \frac{B - B_\infty}{B_w - B_\infty}. \end{aligned} \quad (15)$$

The following are the governing equations after the previously indicated transformation:

$$D_2(f f'' - f \iota) + D_1 f''' - D_3 M f \iota - D_1 K f \iota = 0, \quad (16)$$

$$(Rd + D_4) \theta'' + Pr H (\theta_p - \theta) + Pr f \theta \iota D_5 + Pr A \theta = 0, \quad (17)$$

$$Pr \gamma H (\theta - \theta_p) + \theta_p'' = 0, \quad (18)$$

$$\phi'' + Sc f \phi' - Sc Rc (1 + \delta \theta)^m \exp\left(\frac{-E}{1 + \delta \theta}\right) \phi = 0, \quad (19)$$

$$\Theta'' - pe (\{\Omega + \Theta\} \Psi'' + \Theta' \Psi') + S b f \Theta' = 0, \quad (20)$$

$$\Psi'' + S e f \Psi' - A S e \Theta = 0. \quad (21)$$

The BCs are given:

$$\theta_p(\infty) \rightarrow 0, \theta(0) = 1, f'(0) = 1, \quad (22)$$

$$\begin{aligned}\phi(0) &= 1, \\ f(0) &= 0, \\ \Theta(0) &= 1, \\ \Psi(0) &= 1,\end{aligned}\tag{23}$$

$$\begin{aligned}\theta(\infty) &\rightarrow 0, \\ f'(\infty) &\rightarrow 0, \\ \theta_p(0) &= 1,\end{aligned}\tag{24}$$

$$\begin{aligned}\Psi(\infty) &\rightarrow 0, \\ \Theta(\infty) &\rightarrow 0, \\ \phi(\infty) &\rightarrow 0.\end{aligned}\tag{25}$$

where $Sc = \frac{\nu_f}{D_B}$ Schmidt number, $\Omega = \frac{G_\infty}{(G_w - G_\infty)}$ microorganisms difference parameter, $M = \frac{\sigma_f B_0^2}{\rho_f c}$ magnetic parameter, $Pr = \frac{c_p \mu_f}{k_f}$ Prandtl number, $Sb = \frac{\nu_f}{D_m}$ bio-convection Schmidt number, $pe = \frac{bWc}{D_n}$ Peclet number, $Se = \frac{\nu_f}{D_s}$ Oxytactic microorganism Schmidt number. Interphase heat transfer parameter $H = \frac{\alpha_f h_{fs}}{ck_f}$, Porosity modified ratio parameter $\gamma = \frac{k_f}{lk_s}$.

The subsequent are the numerical formulas for the oxytactic microbe gradient (Nn), solute gradient (Sh) and thermal gradient (Nu) :

$$Nn = \frac{H_3}{D_m(G_w - G_\infty)},\tag{26}$$

$$Nu = \frac{H_1}{k_f(R_w - R_\infty)}, Sh = \frac{H_2}{D_B(S_w - S_\infty)},\tag{27}$$

$$H_1 = -\left(k_{thnf} + \frac{16R_\infty^3 \sigma^*}{3k^*}\right) \frac{\partial R}{\partial y} \Big|_{y=0}, H_2 = -D_B \frac{\partial S}{\partial y} \Big|_{y=0},\tag{28}$$

$$H_3 = -D_m \frac{\partial G}{\partial y} \Big|_{y=0}.\tag{29}$$

$$Sh(Re)^{-0.5} = -\phi'(0), Nu(Re)^{-0.5} = -(D_3 + \frac{4}{3}Rd)\theta'(0),\tag{30}$$

$$Nn(Re)^{-0.5} = -\Theta'(0),\tag{31}$$

$$\begin{aligned}D_3 &= \frac{\sigma_{hnf}}{\sigma_f}, D_4 = \frac{k_{hnf}}{k_f}, D_1 = \frac{\mu_{hnf}}{\mu_f}, \\ D_2 &= \frac{\rho_{hnf}}{\rho_f}, D_5 = \frac{(\rho C_p)_{hnf}}{(\rho C_p)_f}.\end{aligned}\tag{32}$$

Table 1 The characteristics of the improper fluid and nanoparticles [36]

Properties	c_p	k	σ	ρ
Mgo(s_2)	879	30	8×10^{-4}	3580
Ag(s_1)	235	429	6.30×10^7	10490
Water	4179	0.613	0.05	997.1

Xue thermal conductivity model [35] Tables 1 and 2.

$$\frac{k_{nf}}{k_f} = \frac{\ln \frac{k_3+k_f}{2k_f} \left(\frac{k_3}{k_3-k_f} \right) 1 - \varphi_1 + 2\varphi_1}{\ln \frac{k_3+k_f}{2k_f} \left(\frac{k_f}{k_3-k_f} \right) 1 - \varphi_1 + 2\varphi_1}. \quad (33)$$

$$\frac{k_{hnf}}{k_{nf}} = \frac{\ln \frac{k_2+k_{nf}}{2k_{nf}} \left(\frac{k_2}{k_2-k_{nf}} \right) 1 - \varphi_2 + 2\varphi_2}{\ln \frac{k_2+k_{nf}}{2k_{nf}} \left(\frac{k_{nf}}{k_2-k_{nf}} \right) 1 - \varphi_2 + 2\varphi_2}, \quad (34)$$

Table 2 Thermos-physical properties of HN [36]

Characteristics	Hybrid Nanofluid
μ_{hnf}	$\frac{\mu_{hnf}}{\mu_f} = (-\varphi_1 + 1)^{-2.5} (-\varphi_2 + 1)^{-2.5}$
ρ_{hnf}	$\frac{\rho_{hnf}}{\rho_f} = \left[(1 - \varphi_2) + \Phi_2 \frac{\rho_{s2}}{\rho_f} \left\{ \varphi_1 \frac{\rho_{s1}}{\rho_f} + (1 - \varphi_1) \right\} \right]$
σ_{hnf}	$\frac{\sigma_{hnf}}{\sigma_{nf}} \times \frac{\sigma_{nf}}{\sigma_f} = \frac{\sigma_{inf}}{\sigma_f} = \left[\frac{\sigma_{s2} + \sigma_{nf} - \varphi_2 (\sigma_{nf} - \sigma_{s2})}{(\sigma_{nf} - \sigma_{s2}) \varphi_2 + \sigma_{nf} + \sigma_{s2}} \right] \times \left[\frac{\sigma_{s1} + \sigma_f - \varphi_1 (\sigma_f - \sigma_{s1})}{\sigma_f + \varphi_1 (\sigma_f - \sigma_{s1}) + \sigma_{s1}} \right]$
$(\rho c_p)_{hnf}$	$\frac{(\rho c_p)_{hnf}}{(\rho c_p)_f} = \left[(1 - \Phi_2) \left\{ \Phi_1 \frac{(\rho c_p)_{s1}}{(\rho c_p)_f} + (1 - \Phi_1) \right\} + \Phi_2 \frac{(\rho c_p)_{s2}}{(\rho c_p)_f} \right]$

3 Solution methodology

The nonlinear ordinary differential equations (ODEs) presented in Eqs. (16)–(21) are numerically resolved using the bvp4c method with boundary constraints (22)–(25). The original system of PDEs (partial differential equations) is turned into ODEs using similarity variables, which are subsequently transformed into a system of first-order ODEs. For this, the following substitutions are utilized.

$$f' = \beta_2, f'' = \beta_3, f''' = \beta_3 t, \theta = \beta_4, f = \beta_1 \theta' = \beta_5, \theta'' = \beta_5 t, \quad (35)$$

$$\theta_p = \beta_6, \quad \theta'_p = \beta_7, \quad \theta''_p = \beta'_7, \quad \phi = \beta_8, \quad \phi' = \beta_9, \quad \phi'' = \beta'_9, \quad (36)$$

$$\Theta = \beta_{10}, \quad \Theta' = \beta_{11}, \quad \Theta'' = \beta'_{11}, \quad (37)$$

$$\Psi = \beta_{12}, \quad \Psi' = \beta_{13}, \quad \Psi'' = \beta'_{13}, \quad (38)$$

$$\beta'_2 = \beta_3, \quad \beta'_1 = \beta_2, \quad (39)$$

$$\beta'_3 = \frac{1}{D_1} [D_2 (\beta_3 \beta_1 - \beta_2^2) + D_3 M(\beta_2) + D_1 K \beta_2] \quad (40)$$

$$\beta'_4 = \beta_5, \beta'_5 = \frac{1}{\left(\frac{4}{3} Rd + D_4 \right)} (-Pr D_5 (\beta_1 \beta_5) - Pr H (\beta_6 - \beta_4) - APr \beta_4), \quad (41)$$

$$\beta'_6 = \beta_7, \beta'_7 = -Pr \gamma H (\beta_4 - \beta_6), \quad (42)$$

$$\beta'_8 = \beta_9, \beta'_9 = [-Sc (\beta_1 \beta_9) - Sc Rc (1 + \delta \beta_4)^m \exp\left(\frac{-E}{1 + \delta \beta_4}\right) \beta_8], \quad (43)$$

Table 3 Results of machine learning

Case	1	2	3	4	5	6
Performance	$7.92 \times E^{-12}$	$1.32 \times E^{-12}$	$2.49 \times E^{-12}$	$8.29 \times E^{-12}$	$5.61 \times E^{-11}$	$9.81 \times E^{-12}$
Training	$7.87 \times E^{-12}$	$1.28 \times E^{-12}$	$2.49 \times E^{-12}$	$8.04 \times E^{-12}$	$5.69 \times E^{-11}$	$1.00 \times E^{-11}$
Testing	$8.52 \times E^{-12}$	$1.74 \times E^{-12}$	$2.48 \times E^{-12}$	$1.12 \times E^{-11}$	$4.71 \times E^{-11}$	$6.55 \times E^{-12}$

$$\beta'_{10} = \beta_{11}\beta'_{11} = \left[-Sb(\beta_1\beta_9) - Pe(\beta_9\beta_{13} + \{\beta_8 + \Omega\}\beta'_{13}) \right], \quad (44)$$

$$\beta'_{12} = \beta_{13}\beta'_{13} = [-Se(\beta_1\beta_{12}) + \alpha Se\beta_8]. \quad (45)$$

Boundary conditions

$$\beta_2(0) = 1, \beta_1(0) = 0, \beta_8 = 1, \beta_{10} = 1, \beta_{12} = 1, \quad (46)$$

$$\beta_2(\infty) \rightarrow 0, \beta_4(\infty) \rightarrow 0, \beta_{10}(\infty) \rightarrow 0, \beta_{12}(\infty) \rightarrow 0, \quad (47)$$

$$\beta_4(0) = 1, \beta_6(0) = 1, \beta_6(\infty) \rightarrow 0, \beta_8(\infty) \rightarrow 0. \quad (48)$$

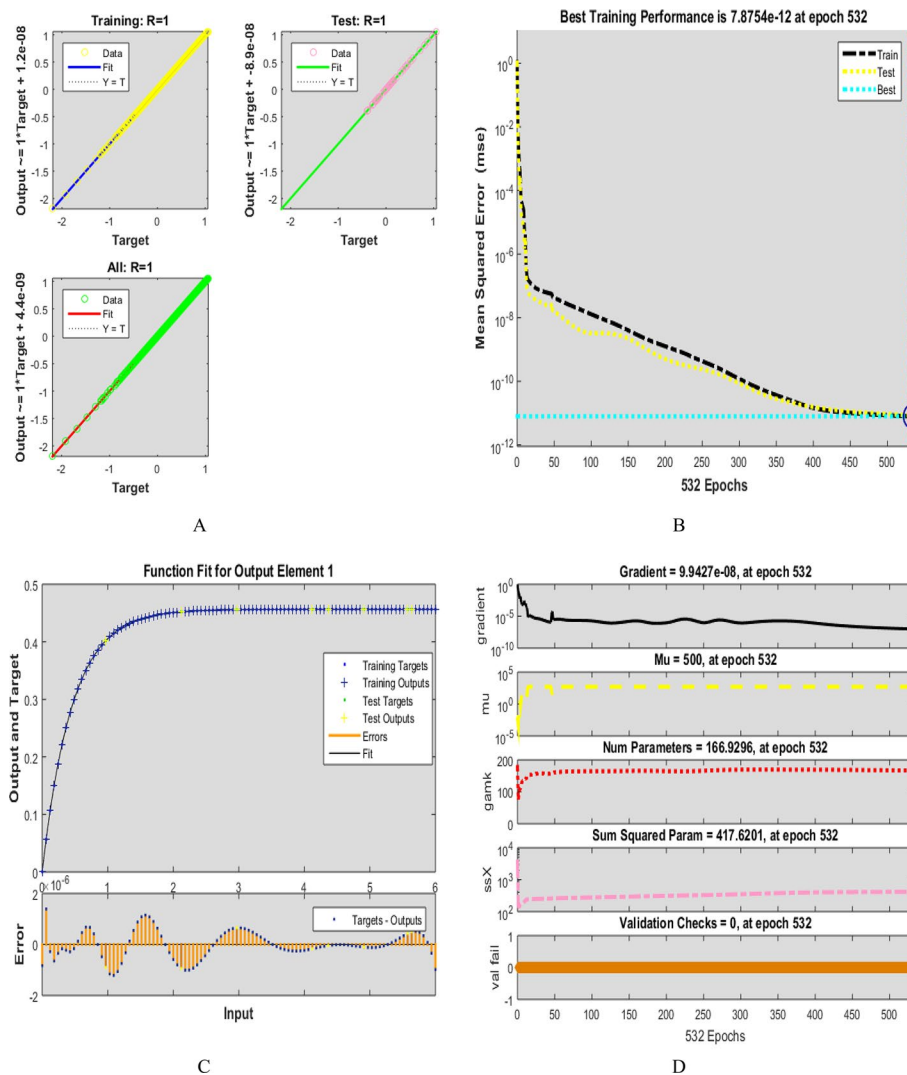


Fig. 2 Case 1

4 Artificial neural network

Machine learning algorithms (MLAs) have developed to the point where they are capable of solving difficult problems in a range of scientific, technical, and mathematical domains, depending on the training strategy and the quality of the available data. Supervised learning algorithms (SLAs) and human learning both depend on dataset analysis to find solutions. In this study, SLAs are utilized to modify network weights to achieve the desired outcomes. A learning signal, also known as a training sample or dataset, is required for neural networks (NNs) in order to guide weight adjustments. By carefully adjusting the NN's regulating weights, the back-propagation process reduces the error that arises when comparing the network's expected output for a particular input with the actual (target) output (Table 3).

The full regression study for the testing and training datasets is shown in Figs. 2A, 3A, 4A, 5A, 6A, and 7A. A regression coefficient ($R = 1$) and virtually perfect arrangement of the regression line with the target data suggest a high degree of correlation between the predicted and actual findings. This demonstrates how well the ANN captures the various physical interdependencies and nonlinear interactions between the regulatory components that affect the heat transfer mechanism. The combined results from Figs. 2A, 3A,

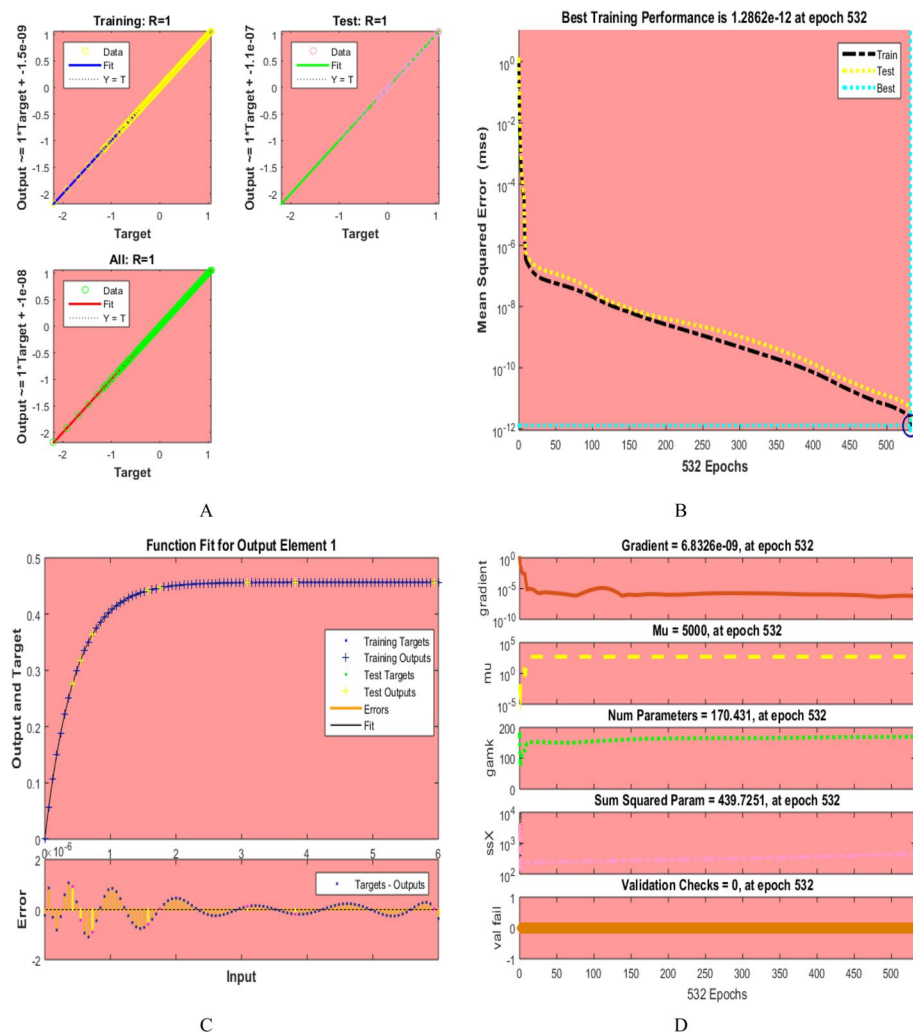


Fig. 3 Case 2

4A, 5A, 6A and 7A show that the Bayesian-Regularization technique provides an incredibly accurate, stable, and dependable computational framework for modeling complex heat transport phenomena when paired with a well-trained ANN model. By guaranteeing greater fidelity and forecast consistency when compared to experimental and numerical reference datasets, regression analysis further improves the model's interpretability. The neural network's remarkable capacity to generalize over unknown data is illustrated in Figs. 2B, 3B, 4B, 5B, 6B and 7B, which show the best training performance at the appropriate epochs for the testing datasets and training. Performance trends show that the network successfully learns the mathematical features of the problem and maintains a good prediction ability outside of the trained data range. The solution fitting graphs shown in Figs. 2C, 3C, 4C, 5C, 6C and 7C further demonstrate the accuracy of the ANN model. Since the projected curves closely match the actual data, it is evident that the ANN has learned the nonlinear mapping between the input and output variables. Better data-fitting qualities and fewer variances follow. The differences in gradient, and training performance for six example scenarios are shown in Figs. 2D, 3D, 4D, 5D, 6D and 7D. These images demonstrate the artificial neural networks (ANN) progressive and adaptive learning behavior as well as its capacity to iteratively eliminate training faults using

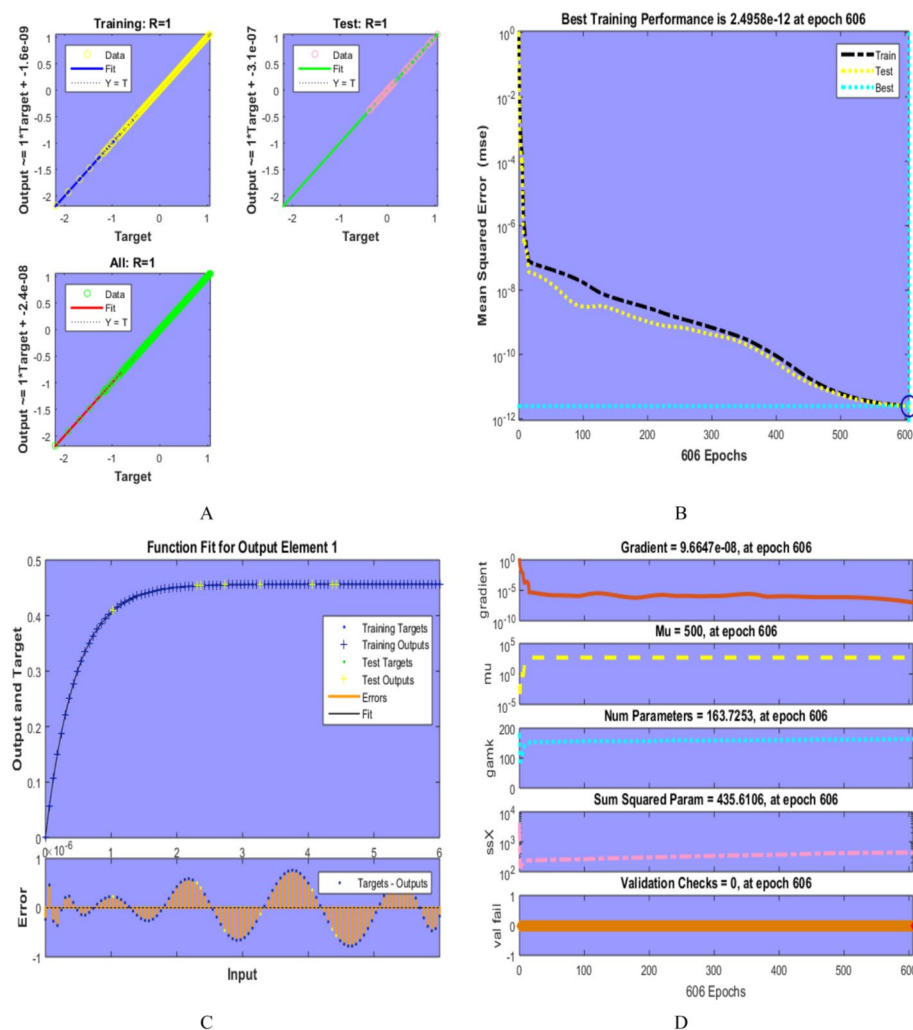
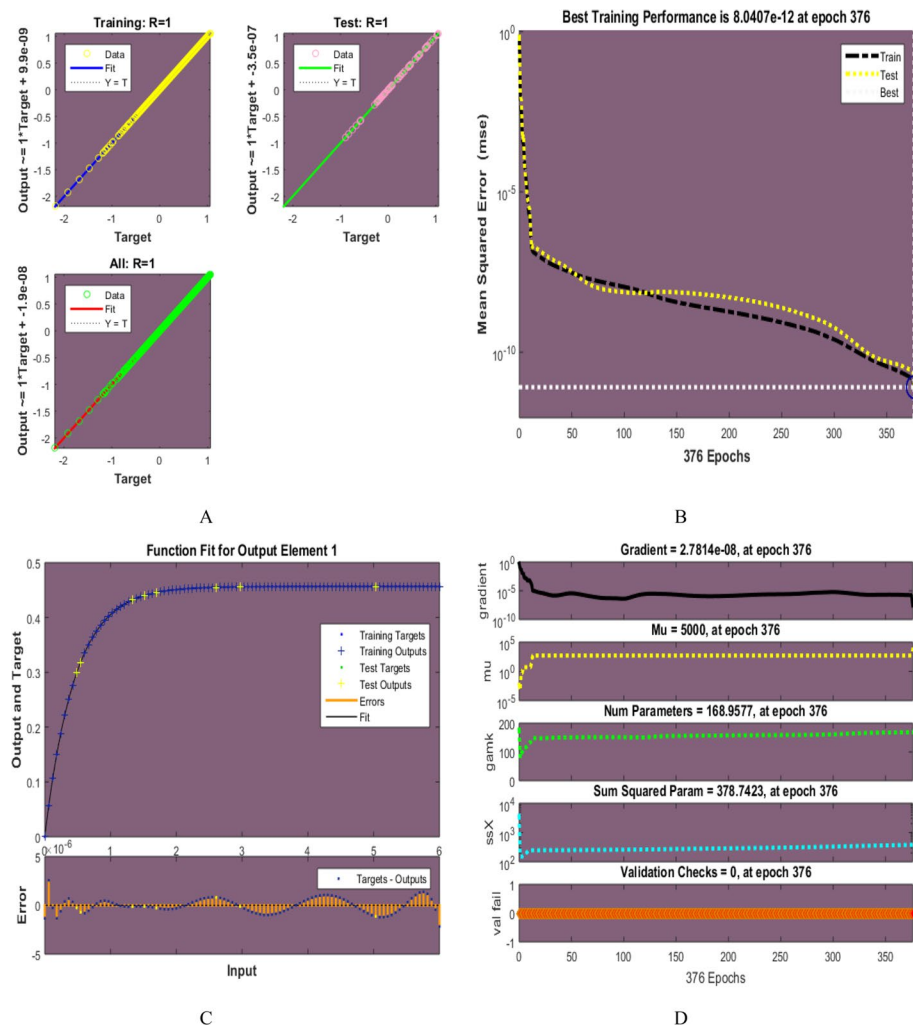


Fig. 4 Case 3

**Fig. 5** Case 4

the Bayesian-Regularization technique. The gradient and error values gradually decrease as the number of epochs rises, suggesting a steady convergence process and verifying the suggested training method's numerical effectiveness. This method offers several benefits for modern thermal-fluid analysis, enabling the optimization of engineering systems with enhanced computational efficiency and physical realism. The error histograms displayed in Fig. 8 reveal the numerical stability and convergence characteristics of the proposed ANN model. There are probably few deviations from the predictions because the majority of the error values are concentrated near zero, suggesting that the network has a high level of precision and accuracy. Such a narrow error distribution illustrates the model's exceptional ability to simplify across many datasets and maintain stability across multiple training cycles.

5 Results and discussion

This model has numerous uses in advanced thermal and bioengineering systems where coupled fluid flow, heat transport, and microbial dynamics are critical. It is applicable to the design of bioreactors and microbial fuel cells, where Oxytactic bacteria enhance mixing and mass transfer efficiency. Figures 2, 3, 4, 5, 6, 7 and 8 shows the Xue thermal

conductivity formulation, the model optimizes heat transport in nanofluid-based cooling systems, notably under magnetic fields (MHD). Figure 9(a&b) display the comparison of ANN, numerical solution and the absolute error plot for $f'(\xi)$ via parameter K . The effects of K on $f'(\xi)$ is display in Fig. 10a. The velocity profile of hybrid nanofluid decline as increase the values of K . The fluid flow decelerates as the resistance increases in proportion to the permeability variable. Figure 10(b) Comparison plot between numerical & ANN and Absolute error plot of parameter Se on Ψ profile. The properties of Se on $\Psi(\xi)$ is shows in Fig. 11a. The Oxytactic microorganism field decline as augments the values the Oxytactic Schmidt number. Figure 11(b) display the comparison plot between numerical & ANN and the absolute error plot of parameter A on θ profile. As rise the values of heat generation factor rise the liquid phase thermal field. The influence of Φ on $f'(\xi)$, liquid and solid phase thermal profiles are display in Figs. 12(a&b) and 13a. The thermal field upsurge as enhances the values of NPs volume friction. This is because the overall fluid motion decelerates down as the solutal of NPs in the liquid beyond the hybrid nanofluid's density threshold. Consequently, the temperature profile rises and velocity decrease. The characteristics of H on $\theta_p(\xi)$ is display in Fig. 13b. The higher H causes the θ_p to decrease. In this case, it is the result of fast heat

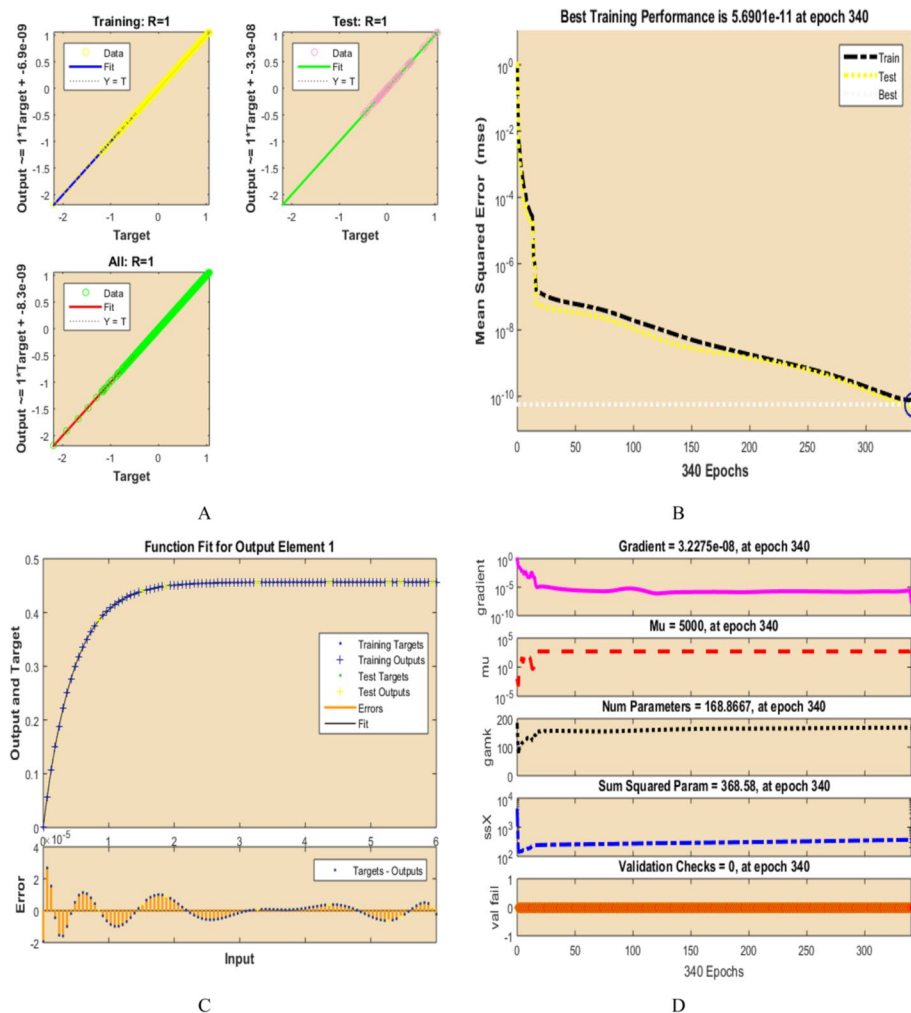


Fig. 6 Case 5

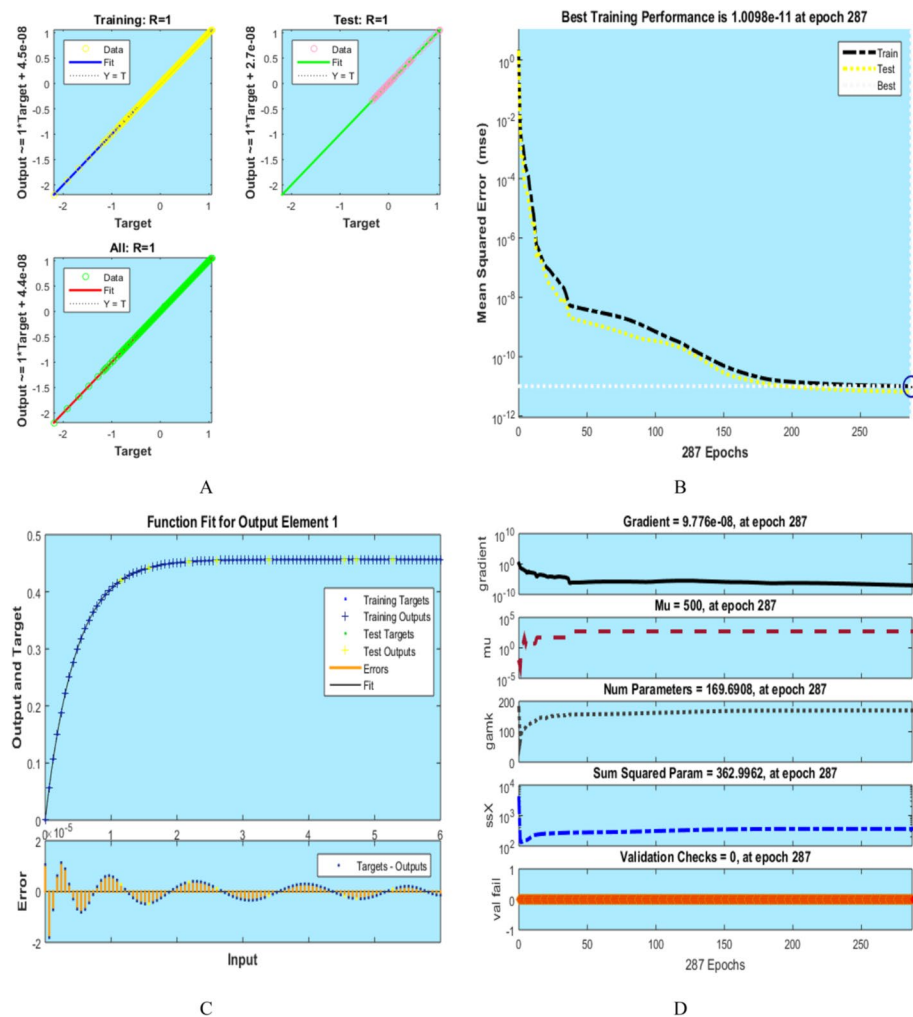


Fig. 7 Case 6

transfer between stages. The upward shift of the parameter H , as shown in the graphs, indicates that process is shifting toward changing the LTE phase. When intermediate values of H are examined, the solid and fluid phases are clearly distinguished, indicating that the flow behavior in this region is governed by. As H increases, the system adjusts to gradually restore thermal equilibrium between the phases. This causes the liquid phase's thermal to progressively get closer to the solid phase's, though the difference may still be noticeable. The system is fully under LTE control when H increases and the heat handover in both phases is equal. The importance of E on $\phi(\xi)$ is display in Fig. 14a. The solutal profile upsurge as rise the values of activation energy factor. Here, a rise in E enhances the solutal gradient. The Arrhenius equation specifically states that the acceleration and heat drop that occur when activation energy is applied to any system lead to a low response rate constant. When the chemical process slows down, the concentration of particles increases. The higher the E , the more the modified Arrhenius mechanism deteriorations. Increasing the nanoparticle concentration is the final catalyst for the producing chemical reaction. The features of pe on $\Theta(\xi)$ is shows in Fig. 14b. As the pe values increased, microbes profile decreased. The pe and W_c are directly proportional to D_n (the diffusivity of microorganisms). The Diffusion and advection rates are affected

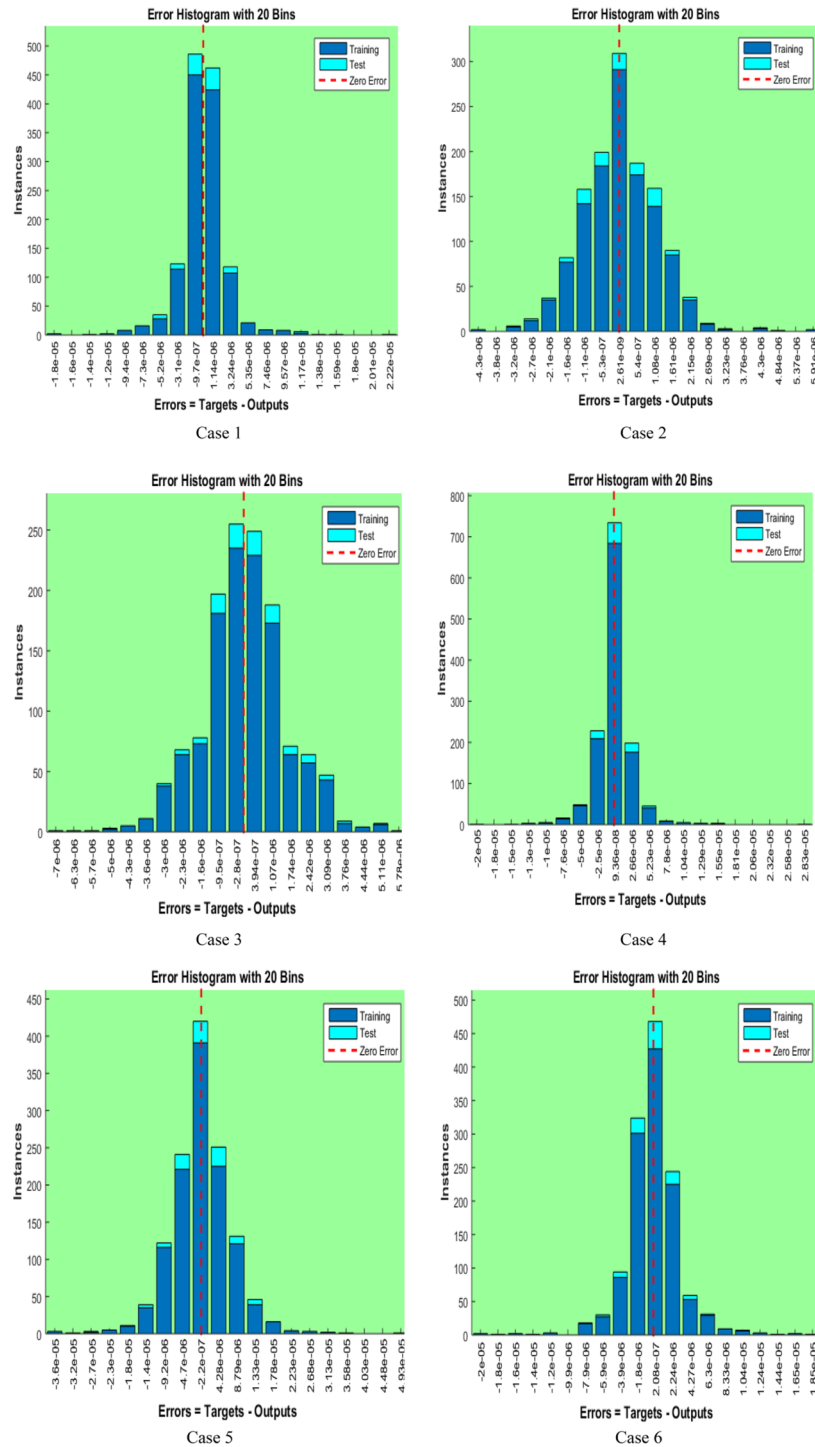


Fig. 8 Error histogram Plots of 6 cases

by the Peclet number. More pe is therefore produced by quicker advective movement, which quickly raises the flow of bacteria. The viscosity of microbes close to the surface is decreased by pe , which makes motile bacteria swim more quickly. The influence of H on Nn and Nu are display in Figs. 15(a&b). As the interphase heat transmission factor values increase, the solid phase heat transmission rate increases and the liquid phase heat transmission rate decreases. The outcome of E on Sh is display in Figs. 16(a). The rate

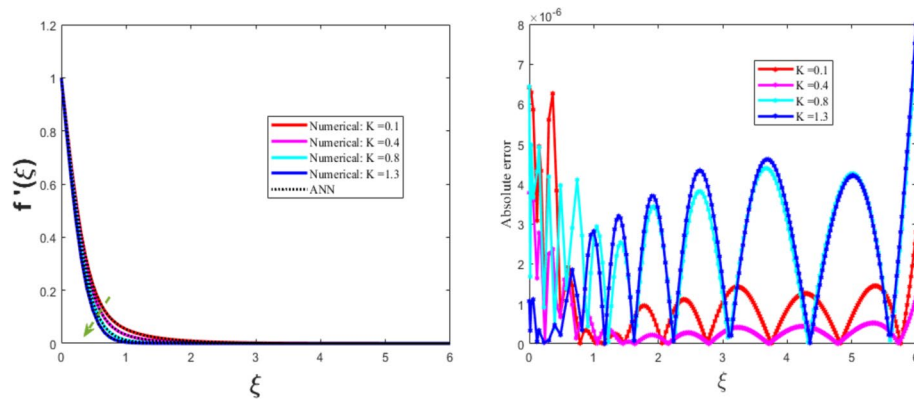


Fig. 9 (a) Comparison plot between numerical & ANN and (b) Absolute error plot of parameter K on F' profile

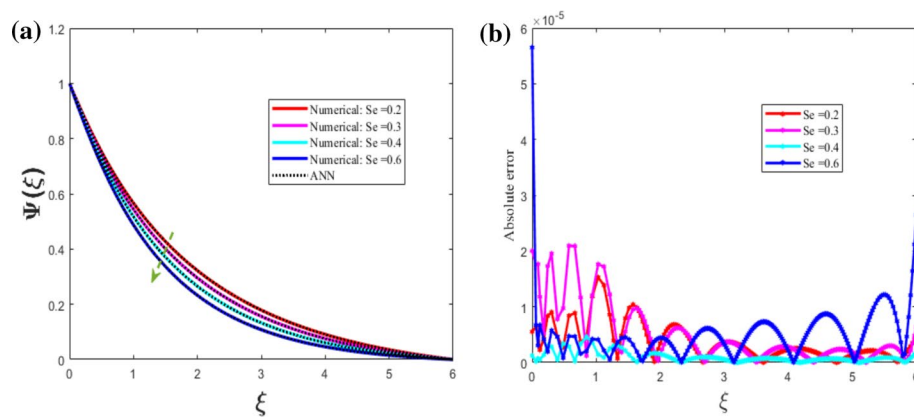


Fig. 10 (a) Comparison plot between numerical & ANN and (b) Absolute error plot of parameter Se on Ψ profile

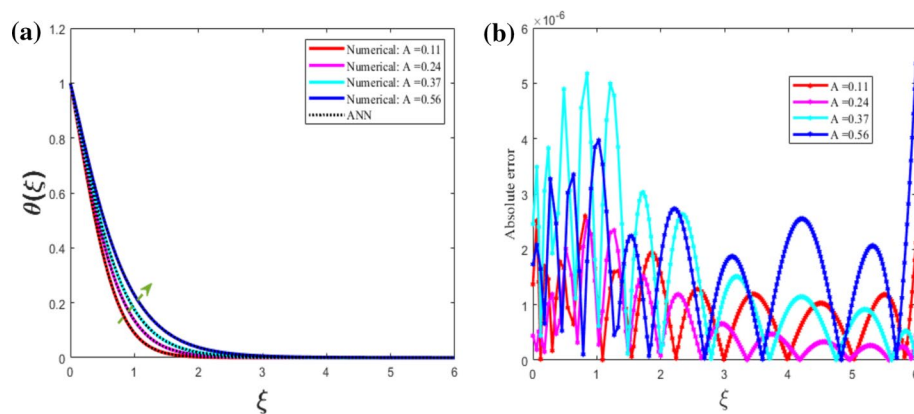


Fig. 11 (a) Comparison plot between numerical & ANN and (b) Absolute error plot of parameter A on θ profile

of mass transmission rise as enhance the values of activation energy factor. The outcome of Sb on Nm is display in Figs. 16(b). As Sb values rise, the local density of microbe declines. To compare the heat transmission rates between the present and published studies, use just the integer case and common factors in Table 4. The two results are very similar, which lends credibility to the current study.

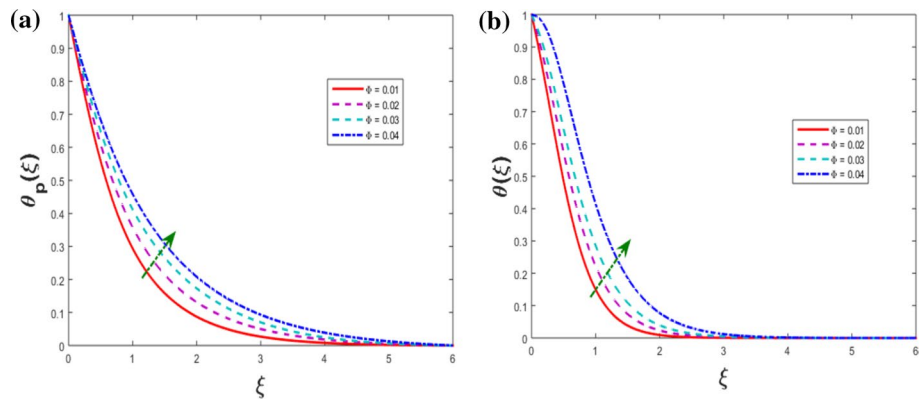


Fig. 12 (a) Significance of Φ on θ_p (b) Properties of Φ on θ

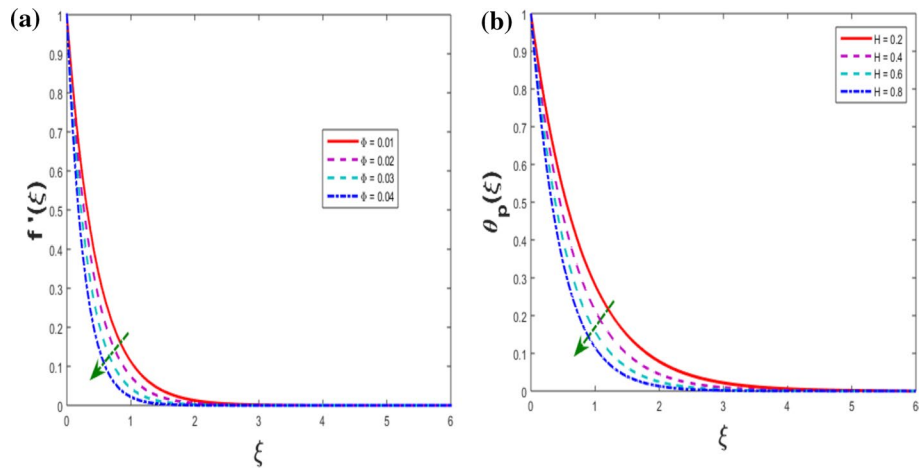


Fig. 13 (a) Outcome of Φ on f' (b) Properties of H on θ_p

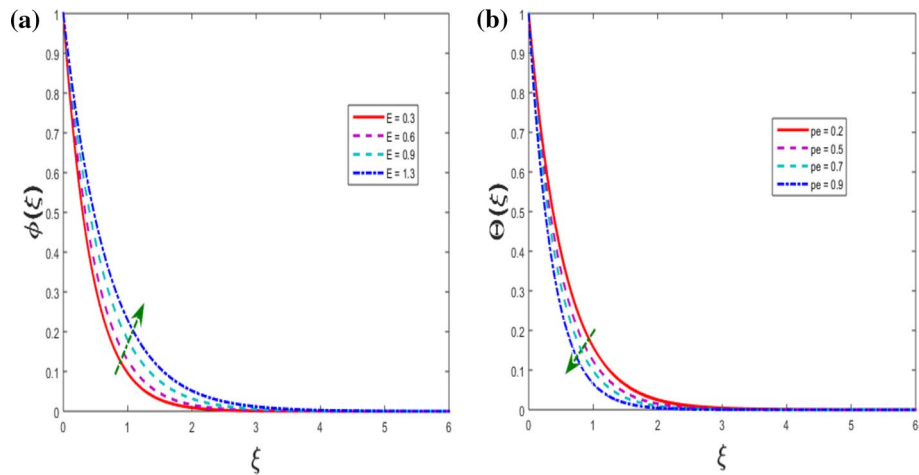


Fig. 14 (a) Result of E on ϕ (b) Features of pe on Θ

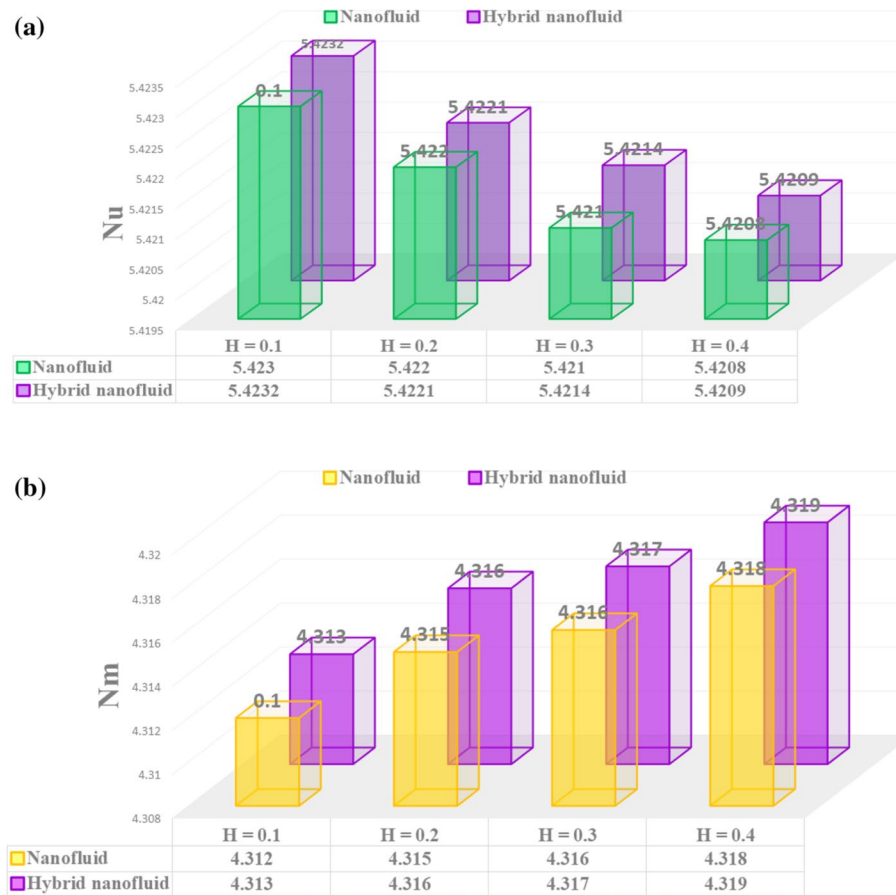


Fig. 15 (a) Properties of H on Nu (b) Properties of H on Nm

6 Conclusions

There are many uses for oxytactic microbes, particularly in environment and industries. Since they actively move in reaction to fluctuations in oxygen level, oxytactic bacteria are frequently used in wastewater treatment for optimal oxygen distribution for the organic materials in accelerators and bioreactors. For a variety of biotechnological applications, oxytactic bacteria must be able to recognize themselves within fluid flows in order to effectively mix and transfer nutrients. The research's primary findings and conclusions are listed below.

- The liquid and solid phase temperature profile increase and the flow profile declines with increasing the nanoparticles volume friction.
- A low mean square error (MSE) indicates the consistency and reliability of the predictions made by the Bayesian Regularized technique.
- The temperature profile increase as rise the values of activation energy parameter.
- The velocity field of the HNF (hybrid nanofluid) is frequently lower as the porosity factor is taken into account.
- The Oxytactic bacteria decline as rise the values of Peclet number.
- Heat transmission in the fluid phase increases when the value of the interphase heat transmission factor rises, while heat transmission in the solid phase upsurges as well.
- The microbe profile tend to decrease with the bioconvection Schmidt number.



Fig. 16 (a) Result of E on Sh (b) Properties of Sb on Nm

Table 4 An examination of the Nusselt number (Nu) for a range of Pr values using published results

Pr	Nu	
	Current results	Punith Gowda et al. [37]
0.71	0.80869	0.80863
1	1.00012	1.00000
3	1.92381	1.92367
10	3.72088	3.72066

Abbreviations

Se	Oxytactic microbe Schmidt number
Nu	Nusselt Number
ρ_f	Fluid density
(x, y)	Cartesian coordinates
k_f	Thermal conductivity ($Wm^{-1}k^{-1}$)
(P, Q)	Velocity fields ($m.s^{-1}$)
pe	Pecklet number
G_∞	Microorganisms
Ω	Microorganism concentration
R_∞	Ambient temperature
σ_f	Electrical conductivity
D_B	Mass diffusivity coefficient (m^2s^{-1})
k_r^*	Chemical reaction
Sh	Sherwood Number

T_m	Liquid temperature
S_∞	Ambient concentration
Nn	Local density of bioconvection
C_p	Specific heat ($Jkg^{-1}k^{-1}$)
E	Activation energy parameter
Sb	Biconvection Schmidt number
B_∞	Ambient Oxytactic microorganisms
ρ_f	Fluid density (kg/m^3)
H	Inter phase heat transfer
ν_f	Kinematic viscosity
Φ	Volume friction of nanoparticles
Pr	Prandtl number
Kn	Local density of Oxytactic microorganism
D_m	Microbes (m^2s^{-1})
γ	Porsity modified conductivity ratio
μ_f	Dynamic viscosity
Sc	Schmidt number
Rc	Chemical reaction parameter

Author contributions

Munawar Abbas and Md. Mahbub Alam: Supervision, Methodology, Conceptualization, Writing – review & editing. Mohamed Bechir Ben Hamida: Methodology, Formal analysis. , Murad Khan Hassani: Writing – review & editing.

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Data availability

All data used in this manuscript have been presented within the article.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent to publish

Not applicable.

Clinical trial

This study is not a clinical trial.

Competing interests

The authors declare no competing interests.

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